SUBHOJYOTI MUKHERJEE

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Research Interests I am a research scientist at Adobe Research working in RL, LLMs/MLLMs, and alignment. In this role, I am engaged in both research, deploying research into Adobe products, and mentoring interns. My research focuses on post-training alignment for LLMs and VLMs, developing methods that make deployed products better, understanding reasoning, synthetic data generation, and reward modeling.

Work Experience **Research Scientist/Engineer, Adobe Inc.**, San Jose, USA 2025, March – present

Products

1. **Pre-training and post-training small LMs (on-device models) for Adobe Document Cloud:**
   - *English Document Overview on-device model in Acrobat Reader (Oct, 2025).* Contributed to pre-training and post-training of the 55mb model. Some of the pre-training data was synthetically generated from GPT-OSS 120B. We used the Nemotron CC pretraining dataset as well.
   - *Spanish Document Overview model in Acrobat Reader.* Led RL post-training for multilingual adaptation. We augmented some of the Spanish data with Spanish synthetic data from SmolLM3-3B.
2. **Post-training VLMs for agentic systems in Adobe Express:**
   - *Express Agent Orchestrator.* Led synthetic data generation and post-training for the tool-use planning system powering the Express AI Assistant (see Adobe MAX 2025 Keynote).
3. **RL post-training for Firefly image/video editing models:**
   - *Image/Video editing.* Contributed to RL post-training of Adobe Firefly Gen5 and Gen6 models. Here I focused on RL post-training. The offline data was generated using rollouts from our existing Firefly models and evaluated on our internal reward models.

Education

University of Wisconsin-Madison, Madison, USA Fall 2019 – 2025, Feb
Ph.D., Electrical & Computer Engineering
Adviser: Dr. Robert Nowak, Dr. Josiah Hanna and Dr. Qiaomin Xie

University of Wisconsin-Madison, Madison, USA Fall 2019 – 2021
M.S, Electrical Engineering
Adviser: Dr. Robert Nowak

Indian Institute of Technology Madras, India 2015–2018
M.S (Research), Computer Science
Advisers: Dr. Balaraman Ravindran and Dr. Nandan Sudarsanam

West Bengal University of Technology, Kolkata, India 2009–2013
Bachelor of Technology, Computer Science & Engineering

Publications

1. Yash Savani, Branislav Kveton, Yuchen Liu, Yilin Wang, Jing Shi, **Subhojyoti Mukherjee**, Nikos Vlassis, Krishna Kumar Singh, *Stepwise Credit Assignment for GRPO on Flow-Matching Models*. **CVPR 2026, main conference**
2. **Subhojyoti Mukherjee**, Viet Dac Lai, et.al. *"Offline RL by Reward-Weighted Fine-Tuning for Conversation Optimization"*. **NeurIPS 2025, main conference** [Paper] (LLM, RL)
3. **Subhojyoti Mukherjee**∗, Yu Xia∗, et.al. *"From Selection to Generation: A Survey of LLM-based Active Learning"*. **ACL 2025, main conference** [Paper] (LLM)
4. Gaurush Hiranandani, Haolun Wu, **Subhojyoti Mukherjee**, Sanmi Koyejo *"Logits are All We Need to Adapt Closed Models"*. **ICML 2025, main conference** [Paper] (LLM, RL)
5. **Subhojyoti Mukherjee**, Josiah Hanna, Qiaomin Xie, Robert Nowak, *"Pretraining Decision Transformers with Reward Prediction for In-Context Multi-task Structured Bandit Learning"*. **RLC 2025** [Paper] (LLM, ICL)
6. **Subhojyoti Mukherjee**, Qiaomin Xie, Robert Nowak, *"Multi-task Representation Learning for Fixed Budget Pure-Exploration in Linear and Bilinear Bandits"*. **RLC 2025** [Paper] (RL)
7. **Subhojyoti Mukherjee**, Anusha Lalitha, Kousha Kalantari, Aniket Anand Deshmukh, Ge Liu, Yifei Ma, Branislav Kveton, *"Optimal Design for Human Preference Elicitation"*. **(NeurIPS 2024, main conference)** [Paper] (LLM, RLHF)
8. **Subhojyoti Mukherjee**, Josiah Hanna, Robert Nowak, *"SaVeR: Optimal Data Collection Strategy for Safe Policy Evaluation in Tabular MDP"*. **(ICML 2024, main conference)**[Paper] (RL)
9. **Subhojyoti Mukherjee**, Anusha Lalitha, Kousha Kalantari, Aniket Anand Deshmukh, Ge Liu, Yifei Ma, Branislav Kveton, *"Optimal Design for K-Way Human Feedback"*. **(Models of Human Feedback for AI Alignment workshop ICML 2024)** [Paper] (LLM, RLHF, OD)
10. Aniruddha Bhargava, Lalit Jain, Branislav Kveton, Ge Liu, **Subhojyoti Mukherjee**, *"Off-Policy Evaluation from Logged Human Feedback"*. **(Models of Human Feedback for AI Alignment workshop ICML 2024)**[Paper] (LLM)
11. **Subhojyoti Mukherjee**, Qiaomin Xie, Josiah Hanna, Robert Nowak, *"SPEED: Optimal Experimental Design for Policy Evaluation in Linear Heteroscedastic Bandits"*. **(AISTATS 2024)**[Paper] (RL, OD)
12. **Subhojyoti Mukherjee**, Qiaomin Xie, Josiah Hanna, Robert Nowak, *"Multi-task Representation Learning for Pure Exploration in Bilinear Bandits"*, Neural Information Processing Systems. **(NeurIPS 2023)** [Paper] (RL, OD)
13. **Subhojyoti Mukherjee**, Josiah Hanna, Robert Nowak, *"ReVar: Strengthening Policy Evaluation via Reduced Variance Sampling"*. Uncertainty in Artificial Intelligence. **(UAI-22)** [Paper] (RL)
14. **Subhojyoti Mukherjee**, *"Safety Aware Changepoint Detection for Piecewise i.i.d. Bandits"*. Uncertainty in Artificial Intelligence **(UAI-22)**.[Paper] (RL)

15. **Subhojyoti Mukherjee***, Ardhendu Tripathy*, Robert Nowak, *"Chernoff Sampling for Active Testing and Extension to Active Regression"*. The 25th International Conference on Artificial Intelligence and Statistics **(AISTATS-22)**. [Paper] (RL, OD)
16. Blake Mason, Romain Camilleri, **Subhojyoti Mukherjee**, Kevin Jamieson, Robert Nowak, Lalit Jain, *"Nearly Optimal Algorithms for Level Set Estimation"*. The 25th International Conference on Artificial Intelligence and Statistics **(AISTATS-22)**. [Paper] (RL, OD)
17. Samarth Gupta, Shreyas Chaudhari, **Subhojyoti Mukherjee**, Gauri Joshi, Osman Yagan, "*A Unified Approach to Translate Classical Bandit Algorithms to the Structured Bandit Setting*", *IEEE International Conference on Acoustics, Speech and Signal Processing **(ICASSP-21)***. [Paper] (RL)
18. Samarth Gupta, Shreyas Chaudhari, **Subhojyoti Mukherjee**, Gauri Joshi, Osman Yagan, "*A Unified Approach to Translate Classical Bandit Algorithms to the Structured Bandit Setting*", *IEEE Journal on Selected Areas in Information Theory **(2020)***. [Paper] (RL)
19. **Subhojyoti Mukherjee**, and Odalric-Ambrym-Maillard, "*Distribution-dependent and Time-uniform Bounds for Piecewise i.i.d Bandits*", *Thirty-sixth International Conference on Machine Learning **(ICML-19)***, Workshop on Reinforcement Learning for Real Life 2019 track [Poster]. [Paper] (RL)
20. **Subhojyoti Mukherjee**, K.P. Naveen, Nandan Sudarsanam, and Balaraman Ravindran, "*Efficient UCBV: An Almost Optimal Algorithm using Variance Estimates*", *Proceedings of the Thirty-Second Association for the Advancement of Artificial Intelligence **(AAAI-18)***, main conference track [Oral].[Paper] (RL)
21. **Subhojyoti Mukherjee**, K.P. Naveen, Nandan Sudarsanam, and Balaraman Ravindran, "*Thresholding Bandits with Augmented UCB*", *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence **(IJCAI-17)***, main conference track [Poster]. [Paper] (RL)

Preprints

1. Branislav Kveton, Anup Rao, **Subhojyoti Mukherjee**, Krishna Kumar Singh, Viet Dac Lai, "AdvantageFlow: Advantage-Weighted Least Squares for RL in Flow Models", [Paper]
2. Daeun Lee, **Subhojyoti Mukherjee**, Branislav Kveton, Ryan A Rossi, Viet Dac Lai, Seunghyun Yoon, Trung Bui, Franck Dernoncourt, Mohit Bansal, "STREAMGAZE: Gaze-Guided Temporal Reasoning and Proactive Understanding in Streaming Videos", [Paper]
3. **Subhojyoti Mukherjee**, Stefano Petrangeli, Branislav Kveton, Trung Bui, Franck Dernoncourt, Arko Mukherjee, *"Agentic Planning with Reasoning for Image Styling via Offline RL"*. [Paper]
4. Puneet Mathur, Branislav Kveton, **Subhojyoti Mukherjee**, Viet Dac Lai, *"Partial Policy Gradients for RL in LLMs"*. [Paper]
5. **Subhojyoti Mukherjee**, Anusha Lalitha, Sailik Sengupta, Aniket Anand Deshmukh, Branislav Kveton, *"Multi-Objective Alignment of Large Language Models Through Hypervolume Maximization"*. [Paper] (LLM, RLHF)
6. **Subhojyoti Mukherjee**, Ge Liu, Aniket Anand Deshmukh, Anusha Lalitha, Yifei Ma, Branislav Kveton, *"Optimal Design for Adaptive In-Context Prompt Tuning in Large Language Models"*. [Paper] (LLM, ICL, OD)

7. **Subhojyoti Mukherjee**, Ruihao Zhu, Branislav Kveton, *"Efficient and Interpretable Bandit Algorithms"*. [Paper] (RL, OD)

Mentorship:

Primary Mentor: Daeun Lee — StreamGaze project (paper submitted to CVPR 2026); Yidhong Huang (current); Alice Mao (current)

Co-Mentor: Junghyun Lee; Yash Savani

Research Internships

1. **Amazon.com Inc., Santa Clara, USA:** Summer 2024 (Full-time), Host: Branislav Kveton, Yifei Ma, Anusha Lalitha, Kousha Kalantiri, Aniket Deshmukh. Working on Alignment for Multi-objective optimization with LLMs.
2. **Amazon.com Inc, Santa Clara, USA:** Fall 2023 (Part-time), Host: Branislav Kveton, Yifei Ma, Anusha Lalitha, Kousha Kalantiri, Ge Liu, Aniket Deshmukh, Anoop Deoras. Working on RLHF with LLMs
3. **Amazon.com Inc, Santa Clara, USA:** Summer 2023 (Full-time), Host: Branislav Kveton, Yifei Ma, Anusha Lalitha, Ge Liu, Aniket Deshmukh, Anoop Deoras. Worked on Active In-Context Learning with LLMs
4. **CMU, ECE Dept., Pittsburgh, USA:** Summer 2019, Host: Gauri Joshi. Worked on Structured Bandits
5. **Adobe Research, San Jose, USA:** Spring 2018. Host: Branislav Kveton. Worked on Item recommendation with Ranking and Bandits
6. **INRIA, SequeL Lab, Lille, France:** Fall 2017, Host: Odalric Maillard. Worked on Non-stationary Bandits

PhD Thesis (ECE, UW-Madison)

Adaptive Data Collection for Policy Evaluation, Multi-Task Learning and LLM Alignment[Thesis] (RL, OD, LLM)

Master's Thesis (EE, UW-Madison)

Active Sequential Hypothesis Testing with Extension to Active Regression and Multi-armed Bandits [Thesis] (RL, OD)

Master's Thesis (CS, IIT Madras)

Finite-time Analysis of Frequentist Strategies for Multi-armed Bandits [Thesis](RL)

Teaching Experience

Teaching/Research Assistant, UW-Madison — 2019–2025
Matrix Methods in Machine Learning - Prof. Robert Nowak
Mathematical Foundation in Machine Learning - Prof. Robert Nowak

Teaching Assistant, UMass Amherst — 2018–2019
Natural Language Processing - Prof. Mohit Iyyer
Design of Algorithms - Prof. Daniel Sheldon

Teaching Assistant, IIT Madras — 2015–2018
Introduction to Programming - Prof. Raghavendra Rao B. V.
Reinforcement Learning(twice) - Prof. Balaraman Ravindran

Reviewer and Service

1. AISTATS, UAI, AAAI, ICML, ICLR, NeurIPS, TMLR, KDD, RLC, CVPR
2. Main Co-ordinator of SILO seminar at UW-Madison

Award Grants and Fellowship

1. Top reviewer award for UAI 2023, Neurips 2023
2. Student Scholarship for AAAI 2018, UAI 2022, Neurips 2023
3. UW-Madison nominee for Apple PhD fellowship and Two-sigma PhD fellowship. Received UW-Madison Chancellor's Opportunity Fellowship 2019-20, UW-Madison ECE Welcome Award of USD 3000.
4. IIT Madras student travel grant of USD 2300, Google travel grant of USD 1700, Microsoft travel grant of USD 1435 (declined).

Other Achievements

Ranked 1150/155190 candidates in Graduate Aptitude Test in Engineering **(GATE)** 2014.
Secured 98.93 percentile in Common Admission Test **(CAT)** 2014 among 196988 candidates.

References

Available Upon Request.